

Michael Vinh Pham

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Education

2021 - 2025	Swarthmore College , B.A., Mathematics (3.7 GPA) & Economics (3.8 GPA) -Sigma Xi Research Society, 2-year D3 Walk-on XC/Track Student-Athlete, Swarthmore College Computer Society
2025 - 2026	New York University , Non-Degree Graduate Program, Computer Science & Economics Coursework: Convex Optimization, PhD Macroeconomics (Dynamic Programming)

Work Experience

2025 - Present (Full-time)	Research , <i>Federal Reserve Bank of New York</i> -Implemented dynamic programming algorithms to solve Heterogeneous Agent New Keynesian (HANK) models -Briefed NY Fed President on r-star and HANK models; worked on DSGE model exhibits reviewed by the FOMC and Chair Jerome Powell -Improved Bayesian posterior estimation: derived higher-order and faster Hessian finite difference equations, stabilized numerical operations for ill-conditioned matrices -Open source contributor to production Julia package DSGE.jl -Led four interns upgrading package compatibility, directing test coverage expansion and reviewing pull requests
Summer 2024 (Full-time) Fall 2024 - Spring 2025 (Part-time)	Economics Research Assistant for Dr. Stephen O'Connell , <i>Swarthmore Economics</i> -Implemented in CUDA dynamic programming algorithms to solve Overlapping generations heterogeneous agent models (finite-horizon stochastic optimal control problem at agent-level w/ equilibrium constraints) (working paper: Human Capital Formation and Inequality in a Heterogeneous Agent Dynastic Model) -Managed research team of three undergraduate researchers
2024 - 2025 (Part-time)	Economics Research Assistant for Dr. Stephanie Kestelman , <i>Harvard Economics</i> -Created ML pipelines (large-scale data scraping, text unsupervised learning, sentiment analysis, LLM fine-tuning) to understand what was being discussed from LA City Planning Commission hearings -Fine-tuned models to perform semantic segmentation on Google Streetview imagery to analyze impact of multi-family housing on urban greenery (working paper: Barking up the Wrong Tree? Multifamily Housing and Tree Loss: Evidence from Los Angeles)
Summer 2023 (Full-time)	Machine Learning Research Assistant for Dr. Ryan Kastner and Dr. Curt Schurgers , <i>UC San Diego CSE (REU)</i> -Researched and implemented computer vision (CV) techniques to semantically segment UAV and satellite imagery, writing PyTorch training scripts for Vision Transformer models to track mangrove deforestation -Created pipeline to source satellite data, perform GIS processing with GeoPandas, feature engineering, and run ML models -Improved lab's mangrove satellite identification from 93% to 98% pixel-wise accuracy
Spring 2023 (Part-time)	Machine Learning Research Assistant for Dr. Xiaodong Qu , <i>Swarthmore CS</i> -Second-authored systematic review of ML for EEG classification (accepted KDD)
Fall 2022- Spring 2023 (Part-time)	Bioinformatics/Machine Learning Research Assistant for Dr. Renuka Nayak , <i>UCSF School of Medicine</i> -Coded DL models in TensorFlow for microbiome phenotype prediction, beating prior SOTA by 30-44pp MCC across three tasks
Summer 2022 (Full-time)	Quantitative Research Intern , <i>Karner Blue Capital</i> -Added Fama-French, CLA Semivariance into firm's Markowitz Portfolio Optimizer, outperforming previous optimizer by 300 bps annually in backtests -Created Flask app for Markowitz Portfolio Optimizer hosted on AWS for internal use by analysts

Personal Projects

July 2026	RLHF: Reward Modeling from Human Preferences - PPO, GRPO to add reward model from preference data on small open source models (Gemma 4 2B)
July 2026	Deep Reinforcement Learning github.com/mpham8/rl-papers - Implemented DQN, PPO, A2C in Python/PyTorch with Pufferlib's Atari, Drone, Cartpole environments
April 2026	Neural Network from Scratch https://github.com/mpham8/nn-fundamental - A no-ML-library basic 2 layer neural network [input (784) → hidden (64) → output (10)] trained with Adam optimizer and mini-batch SGD, with inference implemented in C++.
February 2023	Orange Silicon Valley French-Bambara Machine Translation Challenge - Data pipeline and training of sequence-to-sequence transformer - Orange Silicon Valley Competition 5th Place winner

Personal

Programming Languages	Python, C++/C, Julia
Other	CUDA, PyTorch, Docker
Citizenship	United States
Languages	English (Native), Vietnamese (C1), Spanish (B2), Mandarin (A2)